# Commitment Schemes and Zero-Knowledge Proofs

## CS 171: Cryptography

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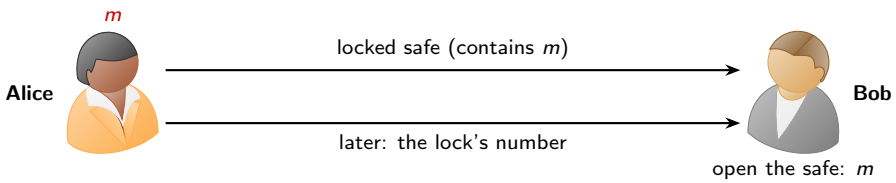
# Overview


1. Commitment Schemes
2. Constructions
3. Statements and Witnesses
4. Zero-Knowledge Proof Systems
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# Commitment

## Goal

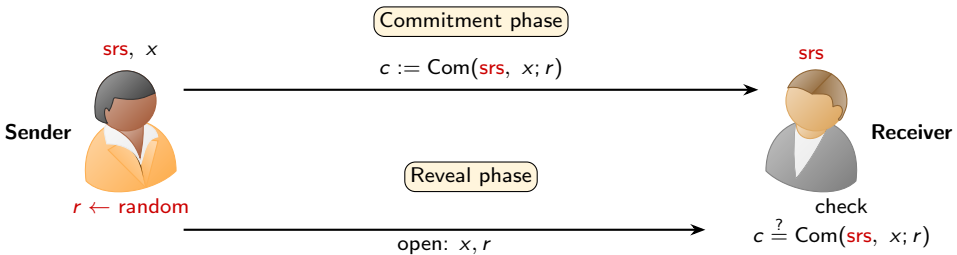
Bind to a secret value that **cannot later be explained with an alternate value** — like locking a message in a safe and handing it over.




**Binding**: cannot open to $m' \neq m$

**Hiding**: $m$ stays secret until then

# Commitment: Two Phases




- Often both parties share a **reference string** $\mathrm{srs} \leftarrow \mathrm{Setup}(1^n)$, e.g. a group and generators.
- In general, each phase can involve several back-and-forth messages. **Correctness**: an honest opening is always accepted.

# Security: Binding and Hiding

## Game $\mathrm{Bind}_{\mathcal{A},\Pi}(n)$

1. $\mathsf{srs} \leftarrow \mathrm{Setup}(1^n)$. $\mathcal{A}$ is given $1^n$ and $\mathsf{srs}$.
2. $\mathcal{A}$ outputs $(x, r)$ and $(x', r')$.
3. Output 1 if $x \neq x'$ and $\mathrm{Com}(\mathsf{srs}, x; r) = \mathrm{Com}(\mathsf{srs}, x'; r')$.

## Game $\mathrm{Hide}_{\mathcal{A},\Pi}(n)$

1. $\mathsf{srs} \leftarrow \mathrm{Setup}(1^n)$. $\mathcal{A}$ is given $1^n$ and $\mathsf{srs}$; it outputs $x_0, x_1$.
2. $b \leftarrow \{0,1\}$, $c := \mathrm{Com}(\mathsf{srs}, x_b; r)$ for a fresh $r$; $\mathcal{A}$ gets $c$, outputs $b'$.
3. Output 1 if $b' = b$.

## Definition 1 (Secure commitment)

$\Pi = (\mathrm{Setup}, \mathrm{Com})$ is **binding** if for all PPT $\mathcal{A}$: $\Pr[\mathrm{Bind}_{\mathcal{A},\Pi}(n) = 1] \leq \mathrm{negl}(n)$, and **hiding** if for all PPT $\mathcal{A}$: $\Pr[\mathrm{Hide}_{\mathcal{A},\Pi}(n) = 1] \leq \frac{1}{2} + \mathrm{negl}(n)$.

- **Binding**: no cheating sender can open one commitment two ways. **Hiding**: $c$ leaks nothing about $x$.

# Perfect or Computational?

## Definition 2 (Hiding)

At the end of the **commitment phase**, no adversarial receiver learns any information about the committed value $x$.

- Against unbounded receivers: **perfectly hiding** ($\Pr[\text{Hide} = 1] = \frac{1}{2}$). Against PPT receivers: **computationally hiding**.

## Definition 3 (Binding)

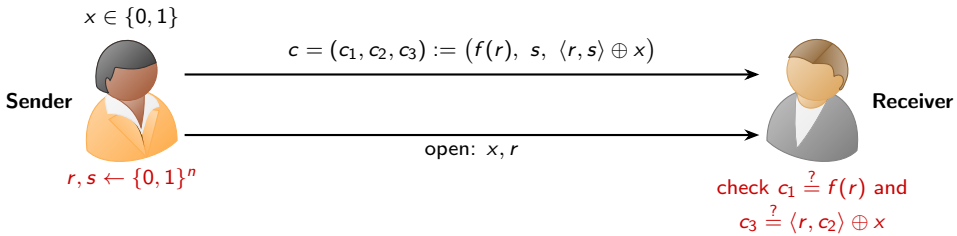
In the **reveal phase**, there is a *unique* value an adversarial sender can reveal as the value committed previously.

- Against unbounded senders: **perfectly binding**. Against PPT senders: **computationally binding**.

- A scheme cannot be **both** perfectly hiding and perfectly binding: if every $x'$ has an opening of $c$, an unbounded sender can find one.

# Commitment from Hardness Concentration

Let $f : \{0,1\}^n \to \{0,1\}^n$ be a one-way **permutation**; commit to one bit $x \in \{0,1\}$. (No srs.)




- **Binding** (perfect): $f$ is a permutation, so $c_1$ determines $r$, and then $x = c_3 \oplus \langle r, c_2\rangle$ is determined.
- **Hiding** (computational): by Goldreich–Levin, $\langle r, s\rangle$ looks like a uniform bit given $f(r) \,\|\, s$ (lecture 12), so $c_3$ is a one-time pad on $x$.

# Can We Use Any Encryption Scheme to Commit?

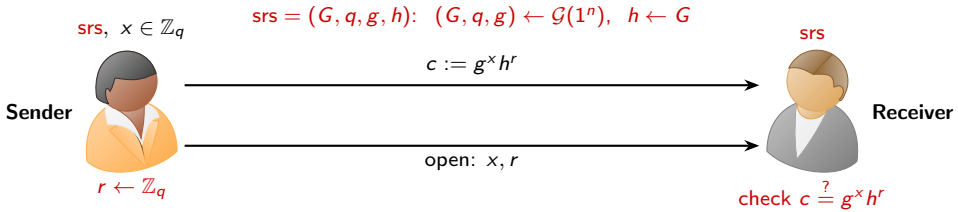
Given $\Pi = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$: let $\mathsf{Com}(x; r)$ use $r$ to run Gen, obtaining $k$, and output $\mathsf{Enc}_k(x)$; open by revealing $r$ (and $x$).

- **No!**
- This commitment is hiding — but it need not be binding.
- Shouldn't binding come from the correctness of decryption? Correctness only holds for honestly chosen keys: a cheating sender may choose its coins dishonestly, e.g. find two keys $k \neq k'$ with $\mathsf{Enc}_k(x) = \mathsf{Enc}_{k'}(x')$.

**Example**

The one-time pad: $c = k \oplus x$ can be opened to *any* $x'$ with the key $k' := c \oplus x'$.

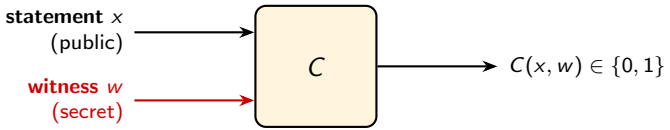
# Pedersen Commitment




- **Binding** (computational): openings $(x, r) \neq (x', r')$ with $g^x h^r = c = g^{x'} h^{r'}$ give $\log_g h = (x - x')/(r' - r) \bmod q$ — the DLog hash of lecture 13 is exactly $(x, r) \mapsto g^x h^r$.
- **Hiding** (perfect): $h^r$ is uniform in $G$, so $c$ is uniform whatever $x$ is. Indeed, for every $c = g^x h^r$ and every $x'$, the value $r' = r + (x - x')/\log_g h$ gives $c = g^{x'} h^{r'}$.

# Statements and Witnesses

A **circuit** $C$: a public, efficient program with two inputs and a one-bit output.



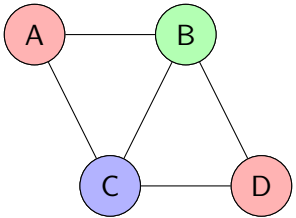

- The statement $x$ is **true** if there *exists* a witness: $\exists w : \ C(x, w) = 1$.
- With $w$ in hand, checking is easy: run $C$. Without $w$, finding one may be hard.

## Examples

- **Discrete log**: $x = (G, q, g, h)$, $w = a \in \mathbb{Z}_q$; $C(x, a) = 1$ iff $g^a = h$. ("I know the secret key for $h$.")
- **Hash preimage**: $x = y$, $w = m$; $C(y, m) = 1$ iff $H(m) = y$.
- **Graph three-coloring**: next slide.

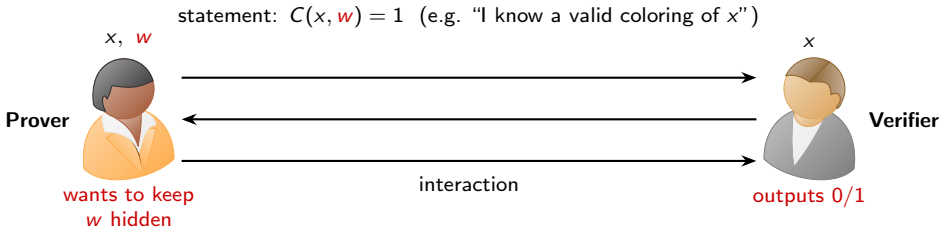
# Graph Three-Coloring

- A graph $G = (V, E)$.
- Task: show a coloring $c : V \to \{R, G, B\}$ such that $c(u) \neq c(v)$ for all $(u, v) \in E$.




- Not every graph is three-colorable, and deciding whether a graph is three-colorable is believed to be computationally hard (it is NP-complete).
- **As a statement**: $x = G$, $w = c$, and $C(G, c) = 1$ iff $c(u) \neq c(v)$ for every $(u, v) \in E$ — just $|E|$ comparisons.

# Zero-Knowledge Proof System




- The verifier should be convinced that $\exists w$ with $C(x, w) = 1$ ...
- ... and learn **nothing else** — in particular, nothing about $w$.

# Zero-Knowledge Proofs: Definition

- **Syntax**: two PPT algorithms, the prover $P(1^n, x, w)$ and the verifier $V(1^n, x)$, which interact; $V$ outputs 0 or 1.
- **Completeness**: an honest prover convinces an honest verifier: for all $x, w$ with $C(x, w) = 1$,

$$\Pr\left[V \text{ outputs } 1 \text{ in } P(1^n, x, w) \leftrightarrow V(1^n, x)\right] = 1 - \mathrm{negl}(n).$$

- **Soundness**: a cheating prover cannot make the verifier accept a **false** statement: for all $P^*$ and all $x$ such that $C(x, w) = 0$ for every $w$,

$$\Pr\left[V \text{ outputs } 1 \text{ in } P^*(1^n, x) \leftrightarrow V(1^n, x)\right] = \mathrm{negl}(n).$$

- **Zero-knowledge**: the interaction leaks nothing about $w$ ... next slide.

# Zero-Knowledge: The Simulator

**Idea**: whatever a (possibly malicious) verifier $V^*$ sees, it could have produced **without** the prover — by running a simulator $\mathcal{S}$ that does not know $w$.

## Game $\mathrm{ZK}_{\mathcal{D},V^*,\mathcal{S}}(n,x,w)$

1. $b \leftarrow \{0,1\}$.
2. If $b=0$: view := the view of $V^*$ in $P(1^n,x,w) \leftrightarrow V^*(1^n,x)$ (real)
   If $b=1$: view $\leftarrow \mathcal{S}^{V^*}(1^n,x)$ (simulated)
3. $\mathcal{D}$ is given $1^n$ and view, and outputs $b'$. Output 1 if $b'=b$.

## Definition 4 (Zero-knowledge)

$(P,V)$ is **zero-knowledge** if there is a PPT simulator $\mathcal{S}$ such that for all PPT $V^*$, all $x,w$ with $C(x,w)=1$, and all PPT $\mathcal{D}$:

$$\Pr\left[\mathrm{ZK}_{\mathcal{D},V^*,\mathcal{S}}(n,x,w)=1\right] \leq \tfrac{1}{2} + \mathrm{negl}(n).$$

# Summary

- **Commitments**: binding and hiding; never both perfect.
- **Constructions**: from a OWP; Pedersen $g^x h^r$.
- **Statements and witnesses**: $x$ is true if $\exists w : C(x, w) = 1$ — e.g. a three-coloring of $G$.
- **Zero-knowledge**: completeness, soundness, and a simulator that fakes the view without $w$.